



HYDROGEN AND ITS COMPOUNDS — FULL MIND MAP

Chemistry · Chapter 6.B · Topic-wise · Fully Explained · Hinglish · SSC/PCS/UPSC Ready

Topics in this chapter:

1. Hydrogen Basics — Valency, Water Composition, Combustion & Storage
2. Heavy Water (D₂O) — Composition, Formula, Reactor Use & Molecular Mass
3. Hard Water, pH & “Purest Water” Concepts
4. Water as a Solvent — Polarity & Dipole Moment
5. Water Purification — Desalination, Chlorination, RO & Alum
6. Hydrogenation of Vegetable Oil — Vanaspati Ghee & Nickel Catalyst

1. Hydrogen Basics — Valency, Water Composition, Combustion & Storage

- Agar element X ke outermost shell mein 4 electrons hain, to uski valency 4 hogi. Hydrogen ki valency 1 hoti hai, isliye compound ka general formula XH₄ hoga — CH₄ jaisa pattern yaad rakho. [Q1]
- Water H₂O ka molar mass 18 g/mol hota hai; isme Hydrogen ka mass 2 g hota hai. Isliye water mein Hydrogen ka mass percentage = $(2/18) \times 100 = 11.11\%$. [Q2]
- 1 mole H₂ ka mass 2 g hota hai. Agar 1 mole H₂ burn hone par 290 kJ heat deta hai, to 4 g H₂ = 2 moles se 580 kJ heat release hogi. [Q3]
- Hydrogen Oxygen ke saath burn karke Water banata hai: $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$. Isi clean combustion-product ki wajah se given fossil-fuel options mein Hydrogen minimum direct air pollution wala fuel maana jata hai. [Q4, Q5]
- Hydrogen ko physically compressed gas ke form mein high-pressure tanks/cylinders mein store kiya ja sakta hai; liquid Hydrogen ke liye cryogenic temperature ($\sim -253^\circ\text{C}$) chahiye. Metal hydrides chemical-storage category mein aate hain. [Q6]

2. Heavy Water (D₂O) — Composition, Formula, Reactor Use & Molecular Mass

- Heavy Water = Deuterium Oxide = D₂O. Deuterium (D) Hydrogen ka isotope hai jisme 1 proton + 1 neutron hota hai; ordinary H₂O ke Protium ko Deuterium replace karta hai. [Q8, Q9, Q11, Q12, Q13]
- PHWR-type nuclear reactors mein Heavy Water neutron moderator bhi ho sakta hai aur coolant bhi. Moderator neutrons ko slow karta hai, aur D₂O ka neutron-absorption relatively low hota hai. [Q7]
- Ordinary water (H₂O) Light-Water Reactors mein moderator + coolant dono roles mein use hota hai, jabki Heavy Water PHWRs mein. Isliye broad “nuclear reactors” wording mein source ne H₂O aur D₂O dono ko valid maana hai. [Q10]
- Heavy Water ka molecular mass ≈ 20 hota hai: 2×2 (Deuterium) + 16 (Oxygen) = 20; ordinary H₂O ka molecular mass 18 hota hai. [Q15]
- Source question Heavy Water discovery ko H. C. Urey se associate karta hai, kyunki Urey ne Deuterium discover kiya aur 1934 Chemistry Nobel jeeta. [Q14]



EXAM TRAP — Q14 mein “Deuterium discovery” aur “pure Heavy Water isolation” ko mix mat karo

- ▶ H. C. Urey ko Deuterium (heavy hydrogen) ki discovery ka credit milta hai.
- ▶ Gilbert N. Lewis ne 1933 mein essentially pure D₂O / heavy-water samples isolate aur study kiye. Isliye historical wording source-key se zyada nuanced hai.

3. Hard Water, pH & “Purest Water” Concepts

- Permanent hardness mainly Calcium aur Magnesium ke chlorides/sulphates ki wajah se hoti hai; temporary hardness mainly bicarbonates se hoti hai. [Q17]
- Pure water neutral hota hai aur 25°C ke around uska pH ≈ 7 hota hai; pH < 7 acidic aur pH > 7 basic/alkaline hota hai. [Q18, Q19]
- Hard water soap ke saath scum banata hai aur boilers mein scale deposit karta hai, isliye washing aur boiler-use mein clearly problematic hai. [Q16]
- Source Q20 “purest form of water” ko Rainwater key karta hai, traditional natural-water context mein; lekin chemistry purity ke sense mein Distilled Water dissolved salts/ions se zyada free hota hai. [Q20]

⚠ EXAM TRAP — Q16 aur Q20 source keys ko blindly generalise mat karo

- ▶ Q16: Hard water drinking aur irrigation ke liye universally “unsuitable” nahi hota. Standard definite problems: soap-washing aur boiler scaling. Given statements ke scientific reading mein 2 & 3 safest hain.
- ▶ Q20: Agar “purest natural water” poochha ho to old exam books Rainwater dete hain; agar option Distilled Water ho aur question simply “purest water” ho, to Distilled Water chemically purer answer hai.

4. Water as a Solvent — Polarity & Dipole Moment

- Water ek polar/dipolar molecule hai aur iska dipole moment high hota hai. O-side partial negative aur H-side partial positive nature ionic species ko attract karti hai. [Q21]
- Isi polarity ki wajah se Water ko “universal solvent” kaha jata hai: ye Na^+ , Cl^- jaise ions ko hydrate/stabilise karke bahut se ionic aur polar substances dissolve kar leta hai. [Q21, Q22]

5. Water Purification — Desalination, Chlorination, RO & Alum

- Saline/impure water se drinking-quality water banane ke broad process ko Desalination kaha jata hai; especially sea water se salts/minerals remove karne ke context mein ye important hai. [Q23]
- Drinking-water disinfection mein Chlorine widely use hota hai; ye bacteria aur dusre microbes ko destroy/inactivate karta hai. [Q24]
- Sea/salty water ko pure water mein convert karne ke liye Reverse Osmosis (RO) use hota hai: osmotic pressure se zyada external pressure laga kar water ko semipermeable membrane ke through reverse direction mein force kiya jata hai. [Q25, Q26]
- Muddy water mein Alum (aluminium sulphate) coagulation karwata hai. Fine charged particles destabilise hoke flocs banate hain aur settle ho jate hain, jisse water clear hota hai. [Q27]

6. Hydrogenation of Vegetable Oil — Vanaspati Ghee & Nickel Catalyst

- Vegetable oil se Vanaspati Ghee banane mein Hydrogen gas add ki jati hai; unsaturated $\text{C}=\text{C}$ bonds hydrogenation ke through zyada saturated fat mein convert hote hain. [Q28]
- Commercial hydrogenation of vegetable oils mein finely divided Nickel (Ni) standard catalyst hota hai. Source approx. $150\text{--}180^\circ\text{C}$ temperature range bhi mention karta hai. [Q29, Q30]
- Exam recall: Vegetable Oil + H_2 —Ni catalyst→ Vanaspati Ghee. Hydrogen reagent hai, Nickel catalyst hai — dono roles alag-alag yaad rakho. [Q28, Q29, Q30]

✓ Source-question mapping preserved: Q1–Q30